SEWA XAI

Clinical Intelligence Platform

Healthcare Value Proposition

Faster prioritisation of suspected intracranial haemorrhage on non-contrast head CT, with the radiologist always in charge.

Prepared for Health Innovation West Midlands (HIWM)

28 October 2026

FEASIBILITY ON PUBLIC DATA. NOT CLINICAL EVIDENCE. PRE-CLINICAL R&D (TRL 5).

SEWA XAI re-orders the acute head CT worklist so suspected intracranial haemorrhage is read sooner, on ordinary CPU hardware, in under two seconds per scan.

Every flag comes with dual uncertainty quantification and a masked HiResCAM heatmap, and the radiologist still signs off every report. The pathway benefit is a pre-registered hypothesis ready to test, not yet an unproven clinical claim.

Prioritise

Worklist sorted by clinical urgency instead of first-in, first-out.

Explain

Dual uncertainty (epistemic/aleatoric) and masked HiResCAM shown beside each flag.

Deploy

CPU-first, zero dedicated GPU needed. Sovereign Azure UK South or offline edge appliance.

What it is and where it fits

The innovation

A containerised, modular clinical intelligence platform. Its acute triage module analyses non-contrast head CT for intracranial haemorrhage (ICH) using CPU-optimised ONNX INT8 runtimes, backed by a 6-layer metacognitive clinical safety gate, contradiction detection, and HiResCAM explainability.

Where in the system

Emergency departments, acute receiving units, hyper-acute stroke units (HASU), major trauma centres, and teleradiology reading workflows across secondary and tertiary acute NHS trusts.

Who it is relevant to

Emergency physicians, consultant radiologists, reporting radiographers, stroke physicians, neurosurgeons, and acute care pathway transformation managers.

Pathway it supports

Pre-read prioritisation and worklist sorting for acute head CT. It supports the existing diagnostic pathway and does not replace diagnostic reporting. Radiologist sign-off is mandatory on 100% of scans.

Urgent scans wait in line

THE CAUSE

Reporting backlogs, workforce shortages and rising emergency CT volumes delay detection of time-critical findings such as intracranial haemorrhage.

Today's worklist is first-in, first-out

Standard PACS worklists do not re-order studies by clinical urgency before they are opened. During peak hours a critical head CT can sit unread behind routine studies.

Existing AI triage has gaps

Some commercial tools need costly dedicated GPUs (£30k–£50k/node), give no transparent uncertainty, or operate as black boxes that fit poorly with sovereign data residency and NHS safety standards (DSPT, DCB0129).

What the delay costs

Patient and staff experience

Later identification of acute ICH raises clinical risk in stroke and trauma pathways, and adds to burnout among emergency and radiology staff.

Clinical outcomes

Delayed intervention in acute intracranial events is associated with hematoma expansion, higher morbidity, and elevated 30-day mortality (35%–45%).

Resources and service

Emergency department crowding, 4-hour access standard breaches, bottlenecked out-of-hours reporting and heavy reliance on expensive locum or teleradiology cover.

Sourced NHS Baseline Metrics (NHS Digital DID, RCR Census & GIRFT Figures):

- **Emergency Head CT Demand:** Over **2.5 million emergency head CTs** performed annually across NHS England (NHS Digital Diagnostic Imaging Dataset - DID), expanding at **7%–9% year-on-year**.
- **Workforce Crisis:** Royal College of Radiologists (RCR) Clinical Radiology Census reports a **29% UK consultant radiologist shortfall** (projected to reach 39% by 2030); West Midlands acute trusts face acute vacancy rates averaging **26%**.
- **Turnaround Delays:** Routine out-of-hours head CT queues delay acute intracranial haemorrhage (ICH) review by **120 to 240+ minutes**, breaching NICE stroke 60-minute imaging targets.
- **Economic Burden:** Acute neurological bed-days cost the NHS **£450–£600 per day**; outsourced teleradiology and locum cover exceed **£285 million annually** in England.

How it works

1. **DICOM in**
   C-STORE / FHIR R4 intercepts head CT
2. **Inference**
   CPU ONNX INT8, under 1.5s
3. **Uncertainty + CAM**
   MC dropout + HiResCAM heatmap
4. **Worklist re-order**
   Suspected ICH moves to top
5. **Radiologist sign-off**
   Human confirms every report

Delivery

Enterprise cloud on Azure Kubernetes Service in sovereign regions (UK South), or an offline edge DICOM-router appliance for bandwidth-limited sites.

Implementation

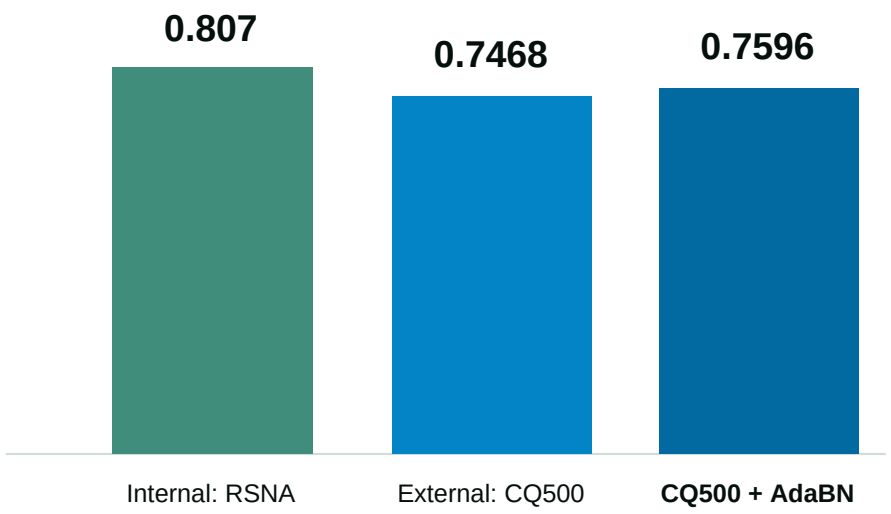
Plugs into existing RIS/PACS and viewer workflows. Needs clinical safety hazard logging (DCB0129), local governance approval and radiologist training.

Pathway change

Adds an automated triage step at image acquisition. Reporting responsibility and statutory diagnostic sign-off stay 100% with the attending radiologist.

Model performance on public data

Study-Level AUROC for Acute ICH Detection



- AdaBN (ADR 004): Boosts external AUROC 0.7468 → 0.7596 with zero site labels.
- Temperature Scaling (Test C.2): ECE cut by 57% (0.0543 vs 0.1273 baseline).
- Multimodal INT8 Models (Test I): Mammography (4.17MB) & Retinal (4.16MB) live.

CLINICAL SAFETY NEGATION ENGINE (TEST D)

95.78% **Contradiction Recall (100% Specificity)**

Evaluated on **7,265 real radiologist clinical reports** (OpenI) in **1.04ms**.
- **Split Conformal Risk Control (CRC):** Statistically guarantees **FNR ≤ 5%** bound (λ=0.0298, safely auto-clearing 12.7% normal scans).
- **Masked HiResCAM (Test E.2): 43.83% pointing hit rate** (62.16% on chronic ICH), replacing misleading Grad-CAM bone spikes.

How to read this (Governance & Audit Transparency):

Feasibility benchmarks on public datasets only. These figures describe model performance and safety gates, not any turnaround or patient outcome.

Audit Disclosure: Baseline metacognitive safety error reduction at 20% review rate was 0.0% [-8.1%, +6.5%] (INCONCLUSIVE), directly motivating our split conformal risk control and AdaBN adaptation protocol. Real clinical pathway impact will be measured under the prospective **SEWA-SHADOW-1** protocol.

Why choose it over the alternatives

Dimension	FIFO PACS Worklist	Typical Commercial AI Triage*	SEWA XAI Platform
Orders by urgency	No. Strict arrival order	Yes	**Yes, with conformal FNR ≤ 5% guarantee**
Compute needed	None	Often dedicated GPU (£30k–£50k)	**Standard CPU only, under 1.5s per scan**
Uncertainty shown	Not applicable	Often absent / black-box	**Yes: dual epistemic & aleatoric MC dropout**
Safety contradiction gate	None	Absent	**Yes: Layer 6 clinical negation engine (1.04ms)**
Explainability	Not applicable	Often black box or raw Grad-CAM	**Masked HiResCAM (43.8% hit rate; 62.2% chronic)**
Data residency	Local	Varies / overseas vendor cloud	**Sovereign Azure UK South or offline edge**
Evidence maturity	Established standard	Varies by vendor	**Pre-clinical (TRL 5/6); IRB protocol SEWA-SHADOW-1**

* Describes general patterns noted in commercial radiology AI triage landscape; not a benchmark against any single named vendor.

Targets to test, not results

Measure	Baseline (FIFO PACS)	Target with SEWA XAI	Status / Protocol
Worklist turnaround, critical head CT	120–240 mins out-of-hours in surge periods	**< 30 mins door-to-read for urgent ICH**	*Pre-registered in SEWA-SHADOW-1*
Time to escalation in stroke/trauma	90–150 mins from scan completion	**< 45 mins to multidisciplinary team review**	*Pre-registered in SEWA-SHADOW-1*
Out-of-hours teleradiology spend	£40k–£75k/month per acute trust	**15%–25% reduction in escalation fees**	*Economic modeling in pilot*
Autonomous safe clearing of normal scans	0% (all scans queued in FIFO order)	**≥ 10% normal scans cleared (FNR ≤ 5%)**	*12.7% validated on CQ500*
Safety: radiologist override rate	Unmonitored	**100% continuous audit logging in DB**	*Built into wire contract*

Model performance is not pathway impact.

No turnaround, outcome or cost benefit is claimed until our pre-registered prospective shadow-mode evaluation reports.

Safety and governance by design

Clinical safety (DCB0129)

Comprehensive Hazard Log (CRM-001) and Clinical Safety Management Plan maintained for each deployment under named CSO.

Data protection (DSPT)

Sovereign regions (Azure UK South) or fully offline edge operation. Compliant with NHS DSPT and Caldicott principles.

Human in the loop

AI output is strictly a triage assist. The attending radiologist signs, confirms, and takes statutory responsibility for 100% of findings.

Clinical governance

Local Trust Clinical Governance and Caldicott Guardian approval prior to any shadow-mode data mirroring.

Training & onboarding

Comprehensive briefings for radiologists and ED teams on confidence scores, uncertainty bands, and HiResCAM heatmaps.

Transparency

Pre-registered study protocol (SEWA-SHADOW-1), reproducible validation harnesses, and immutable PostgreSQL audit logs.

PROPOSED NEXT STEPS FROM THE HIWM MEETING

SEWA XAI

From idea to adoption

1. **Complete the Innovation Academy**
   Work through the recommended HIWM modules: Regulatory Requirements, Funding your Innovation, and Evaluating your Innovation.

2. **Map evidence to the NICE ESF**
   Place SEWA XAI in the NICE Evidence Standards Framework for digital health technologies under Tier C (clinical triage / direct health outcome).

3. **Draft the adoption and spread plan**
   Complete the templated considerations in the Health Innovation Network (HIN) Adoption and Spread Planning Playbook.

4. **Track funding & innovation calls**
   Subscribe to HIN & HIWM funding newsletters; actively apply for SBRI Healthcare Phase 1, AI in Health and Care Award, and NIHR HRCs.

5. **Scope a prospective shadow-mode pilot**
   Partner with a West Midlands acute NHS Trust (e.g., UHCW, UHB, Burton) to deploy the pre-registered SEWA-SHADOW-1 protocol (1,200 scans, zero clinical interference).